Suman Maiti

Email: suman.kgp1999@gmail.com
Mobile: +91-7427960224

Education

- **Indian Institute of Technology Kharagpur** — Kharagpur, India
  Doctor of Philosophy (PhD) - Computer Science and Engineering|Supv Dr.Soumyajit Dey. — *January 2022 - Present*
  Broad area of research: *Security of cyber physical systems.*
  Courses: *High Performance Computer Architecture, Foundations of Algorithm Design and Machine Learning, Artificial Intelligence For Cyber Physical Systems, Computational Foundations of Cyber Physical Systems, Statistical Methods, English for Technical Writing.*
- **Kalyani Government Engineering College** — Nadia, India
  Bachelor of Technology - Electrical Engineering; DGPA: 9.03/10 — *September 2017 - August 2021*
  Courses: *Digital Electronics, Analog Electronics, Control Systems, Signals and Systems, Power Systems, Power Electronics*
- **D.A.V. Model School IIT Campus Kharagpur** — Kharagpur, India
  All India Senior School Certificate Examination, Class 12; Percentage: 93.8% — *March 2017*
  Courses: *Physics, Chemistry, Mathematics, Computer Science, English*
- **D.A.V. Model School IIT Campus Kharagpur** — Kharagpur, India
  All India Secondary School Examination, Class 10; CGPA: 10/10 — *March 2015*
  Courses: *Mathematics, Science, Social Science, Computer Science, English.*

Skills Summary

- **Languages**: Python, PHP, C++, C, Latex
- **Frameworks**: Scikit,TensorFlow, Keras
- **Tools**: Matlab, Simulink, RT Lab, Labview, Artemis, Hyperim, Ephasorsim, Emegasim
- **Platforms**: Linux, Windows
- **Hardwares used**: OP4510 (Opal-rt), SEL-421 (Schweitzer Engineering Laboratories), Raspberry-pi
- **Subjects**: Power System Studies, Control System, Algorithms and Machine Learning
- **Soft Skills**: Event Management, Writing, Public Speaking, Time Management

Experience

- **Teaching assistantship | Guidance: Professor Pallab Dasgupta**
  Department of Computer Science and Engineering, IIT Kharagpur — *March 2023 - Present*
  - **Programming and Data Structures Lab (Section 8)**: As a Teaching Assistant, I assisted students in programming and checked their lab copies for the Programming and Data Structures course at CSE, IIT Kharagpur.
- **Teaching assistantship | Guidance: Professor Soumyakanti Ghosh, Prof. K.S Rao**
  Department of Computer Science and Engineering, IIT Kharagpur — *October 2022 - February 2023*
  - **Programming and Data Structures Lab (Section 18)**: In my role as a Teaching Assistant at CSE, IIT Kharagpur, I assisted students in programming, prepared questions for their assignments, mid-semester exams, and end-semester exams, and checked their lab copies for the Programming and Data Structures course.
- **Teaching assistantship | Guidance: Professor Debasis Samanta**
  Department of Computer Science and Engineering, IIT Kharagpur — *April 2022 - July 2022*
  - **Programming and Data Structures Lab (Section 9)**: As a Teaching Assistant for the Programming and Data Structures course at CSE, IIT Kharagpur, I supported students in programming, conducted viva exams, and graded their lab copies.
- **Junior Research Fellow | Guidance: Professor Soumyajit Dey**
  Department of Computer Science and Engineering, IIT Kharagpur — *October 2021 - Present*
  - **Smart Grid Privacy and Security**: As a Junior Research Fellow at the Department of Computer Science and Engineering, IIT Kharagpur, I conducted research to explore vulnerabilities in smart grid environments and their countermeasures. My work included applying machine learning techniques, particularly reinforcement learning, to identify vulnerabilities in power grid model architectures such as IEEE 9, 14, 37, and 39 bus models.
- **Summer Intern | Guidance: Professor Pabitra Mitra**
  Department of Computer Science and Engineering, IIT Kharagpur — *June 2019 - July 2019*
  - **Design of digital clock using 8085 microprocessor**: During my summer internship in 2019 in the Department of Computer Science and Engineering at IIT Kharagpur, I designed a digital clock using 8085 microprocessor programming. The project involved understanding the working of the 8085 microprocessor and implementing it to display the time using a seven-segment display.

Projects

- **Exploring Formal Methods for Smart Grid Privacy and Security**: Investigating the vulnerabilities of IEEE (9, 14 and 39) bus power grid generation and distribution units with the help of reinforcement learning agent and formal tool Staliro.
- **Fault Studies of Wind Farm Embedded Power Network**: Simulation of symmetrical and asymmetrical fault conditions in a 225/90 *Kilo Volt* power network. Analysis of voltage and power wave forms of the transmission lines during the fault condition.
- **Performance estimation of different CPU architectures using GEM5 system simulator**: Generation of different configuration combinations based on the parameters and their values mentioned by the user, running appropriate benchmark program.
- **Autonomous Driving in F-1/10 vehicle (Work in progress)**: Design of a controller that helps a F-1/10 vehicle to follow a given track while avoiding the obstacles.

Research and Publications

- **Symmetrical and Asymmetrical Fault Studies of Wind-Farm Embedded Power Network**: 2021 Devices for Integrated Circuit (DevIC). IEEE; 2021. Authors: Suman Maiti, Suvodip Som, Soumen Mondal, Pritam Kumar Gayen
- **Work in Progress: A Learning-Assisted Method for Power Grid Vulnerabilities**: 60^{th} Design Automation Conference (DAC) 2023 Authors: Suman Maiti, Anjana B, Sunandan Adhikary, Ipsita Koley, Soumyajit Dey
- **Privacy of real time pricing in smart grids**: IEEE Transanctions on Smart Grid (currently under review)

Test Score

- Qualified Graduate Aptitude Test in Engineering (GATE) in Electrical Engineering, 2021 with score 494.